

Survey Microdata Intelligence & Quality Audit Report

Automated statistical quality assessment, multi-factor anomaly diagnostics, and supervisory compliance report for active survey dataset `generic_survey_2026.csv`.

Dataset File: generic_survey_2026.csv	Dataset ID: 273c7093	Audit Date: 16-Aug-2026 12:22 UTC
Total Sample: 12 Records	Total Variables: 8 Attributes	Audit Status: COMPLETED (PASS)

1. Executive Summary & Quality Scorecards

TOTAL SURVEY RECORDS 12 100% Evaluated by AI Engine	DATA COMPLETENESS RATE 100.0% 0.00% Missing Attribute Rate	HIGH-RISK FLAGGED RECORDS 1 8.33% Priority Review Index	ISOLATION FOREST ANOMALIES 2 16.67% Multivariate Outliers
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Audit Findings Synthesis: The survey microdata dataset (`generic_survey_2026.csv`) underwent comprehensive multi-tier data verification across 12 household observations, representing nationwide microdata sample responses. The dataset demonstrates an overall data integrity completeness score of **100.0%**. A total of **2 deterministic rule discrepancies** and **2 multi-dimensional Isolation Forest outliers** were flagged for supervisory investigation. Overall risk assessment categorizes **10 records (83.3%)** as normal, **1 records** as medium caution, and **1 records** as high-priority audit items.

2. Microdata Statistical Profiling & Key Measures

Variable	Mean	Median	Min – Max	IQR Range (Q1 - Q3)	Skew	Outliers (1.5x IQR)
respondent_age	37.33	36.50	14.0 – 62.0	28.8 – 46.2	0.10	0 (0%)
weekly_hours	38.92	41.00	0.0 – 120.0	28.5 – 45.8	1.17	4 (33.33%)
monthly_income	29,416.67	28,500.00	0.0 – 120,000.0	13,250.0 – 31,250.0	2.46	1 (8.33%)
record_id	106.50	106.50	101.0 – 112.0	103.8 – 109.2	0.00	0 (0%)

4. Geographic & Regional Distribution Highlights

State / Region	Sample Records	National Share	Mean Expenditure (₹)	Avg Household Size
State #Kerala	6	50.00%	■35.67	N/A
State #Tamil Nadu	6	50.00%	■42.17	N/A

5. AI Quality Diagnostics & Priority Flagged Records

Records flagged below combine deterministic business rules (35%), Isolation Forest unsupervised ML anomalies (35%), enumerator skew (15%), and spatial cluster variance (15%):

Record #	Risk Score	Level	Rule Violation	ML Anomaly	Primary Deviation Detail
#4	85/100	High	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#7	50/100	Medium	PASS	OUTLIER	Unsupervised multidimensional anomaly

6. Supervisory Recommendations & Compliance Directives

• **Priority Triage:** 1 high-risk observations require formal supervisor review via the validation queue prior to national statistical aggregation.

- **Integrity Corrections:** 2 rule violations involve field boundary checks (negative income, invalid codes, or missing values) and should be reconciled.
- **Statistical Outliers:** 2 multivariate Isolation Forest anomalies require analytical spot-checking for abnormal expenditure-to-income ratios.

CERTIFIED BY DATA SUPERVISOR	ENDORSED BY DIRECTOR / HEAD OF OPERATIONS
Dr. Rajesh Kumar Lead Statistical Quality Officer Survey Validation Division, MoSPI	Director General (Surveys) National Statistical Office (NSO) Government of India